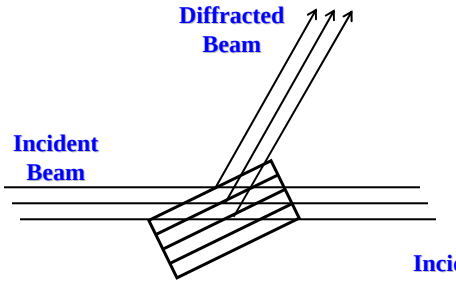
X-ray Powder Diffraction II
Peak Positions & Indexing

Chemistry 754
Solid State Chemistry
Lecture #10

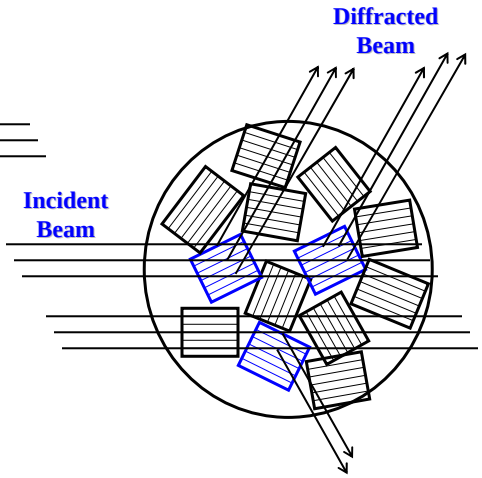
Outline

- Single crystal & powder diffraction
- Information contained in a diffraction pattern
- Geometrical considerations
- Peak positions
- Indexing cubic powder patterns
- Auto indexing

Single Crystal Diffraction

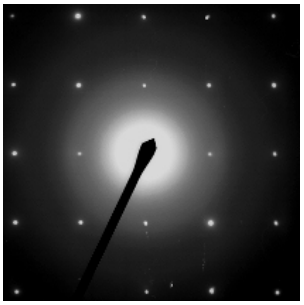


Powder Diffraction

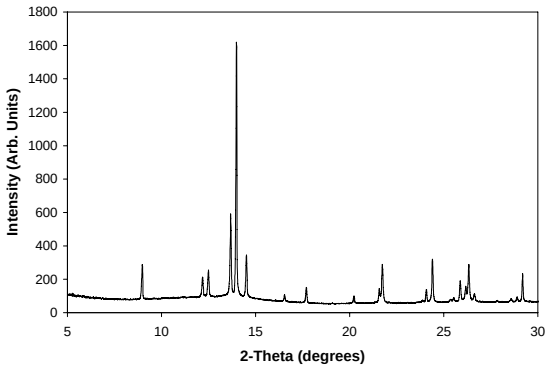


In powder diffraction only a small fraction of the crystals (shown in blue) are correctly oriented to diffract.

Single Crystal Diffraction (3D)



Powder Diffraction (1D)



Single Crystal Diffractometer (4 circles)





Powder Diffractometer (2 circles)

Powder Diffraction Methods

- Qualitative Analysis
  - Phase Identification
- Quantitative Analysis
  - Lattice Parameter Determination
  - Phase Fraction Analysis
- Structure Refinement
  - Rietveld Methods
- Structure Solution
  - Reciprocal Space Methods
  - Real Space Methods
- Peak Shape Analysis
  - Crystallite Size Distribution
  - Microstrain Analysis
  - Extended Defect Concentration

Information in a Diffraction Pattern

- **Peak Positions**
  - The peak positions tell you about the translational symmetry. Namely what is the size and shape of the unit cell.
- **Peak Intensities**
  - The peak intensities tell you about the electron density inside the unit cell. Namely where the atoms are located.
- **Peak Shapes & Widths**
  - The peak widths and shapes can give information on deviations from a perfect crystal. You can learn about the crystallite size if it is less than roughly 100 – 200 nm. You can also learn about extended defects and microstrain.

Debye-Scherrer Camera & Cones of Diffraction

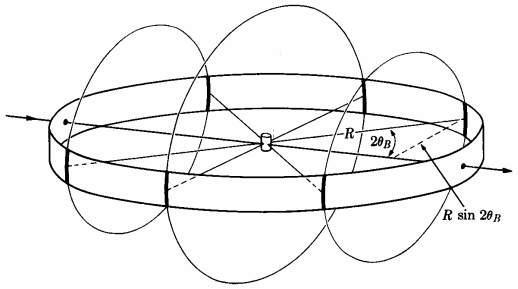
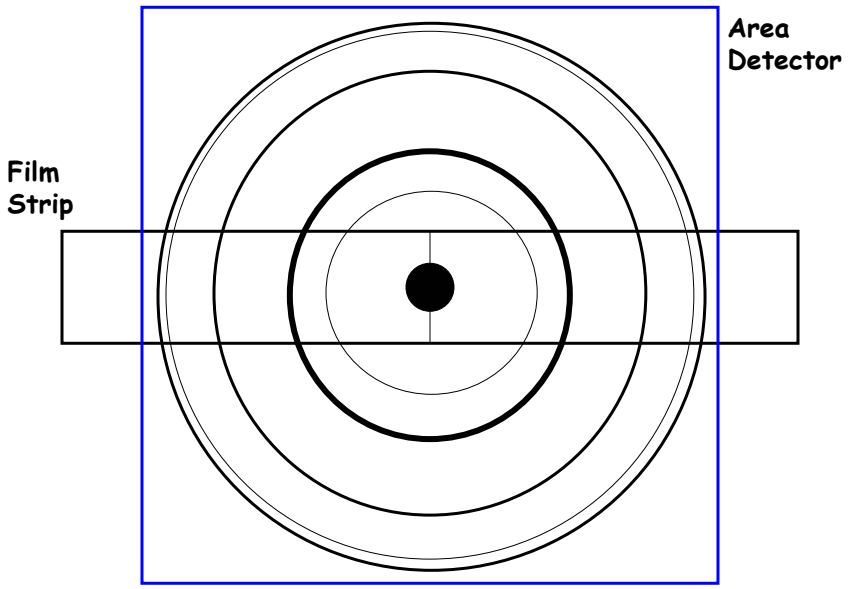


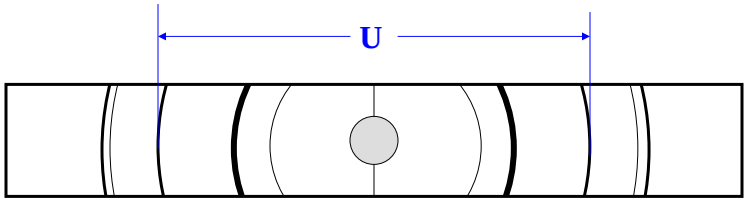
Fig. 4–16 Intersection of cones of diffracted rays with Debye-Scherrer film.

$$\lambda = 2d_{hkl} \sin \theta_{hkl}$$

Measuring Diffraction Rings



Measuring Diffraction Rings

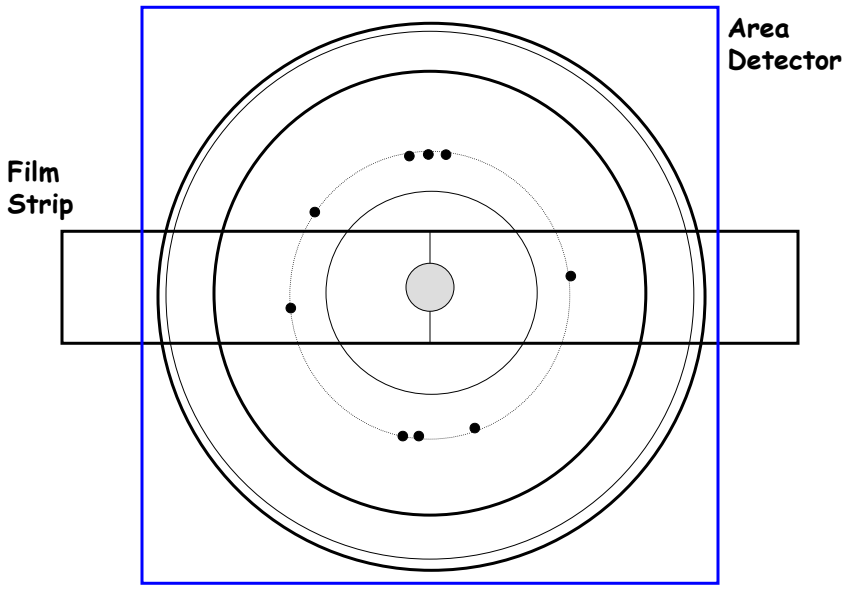


$$U = 4\theta_{hkl} R$$

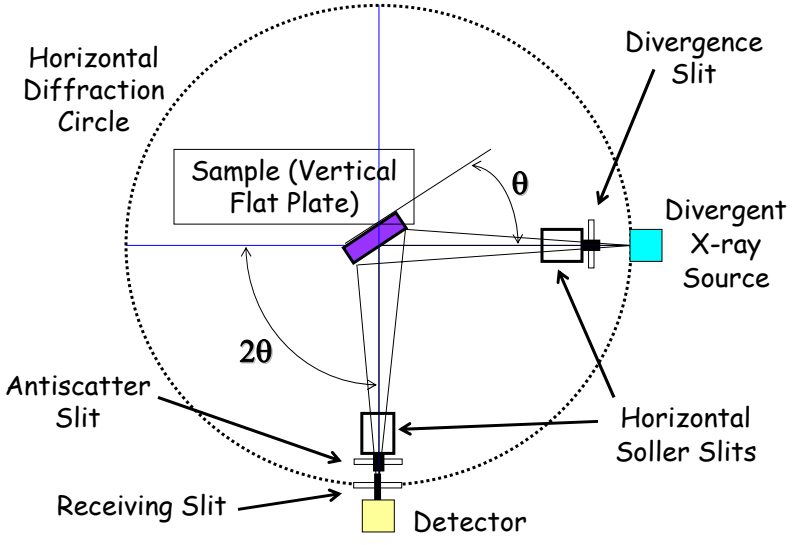
θ_{hkl} = Bragg Angle

R = Diffractometer Radius

Preferred Orientation/Texture



Bragg-Brentano Parafocusing Diffractometer



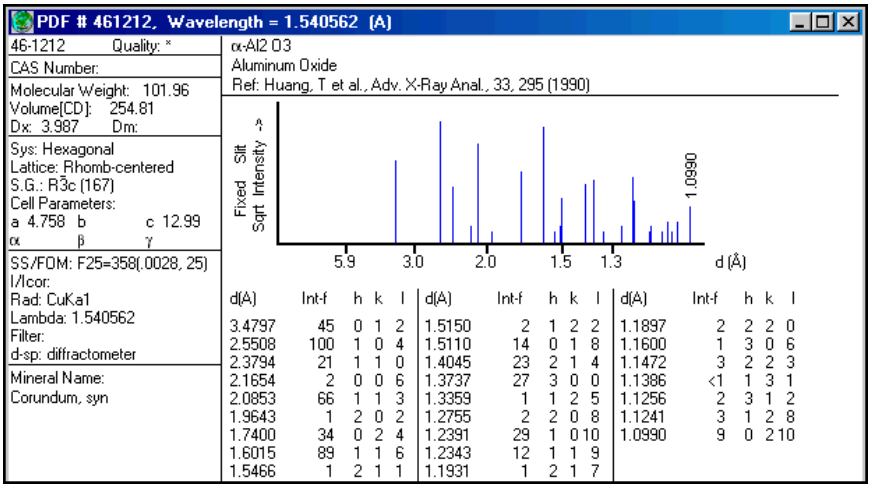
Qualitative Analysis
Searching with the ICDD

Once you have a powder pattern you can use it like a fingerprint to see if it matches the powder pattern of an already known compound. Nowadays this is usually done with the help of a computer.

The International Centre for Diffraction Data (ICDD) maintains a database of known powder diffraction patterns (www.icdd.com)

- 115,000 patterns (not all unique)
- 95,000 Inorganic compounds
- 20,000 Organic compounds

Sample ICDD Card



PDF # 461212, Wavelength = 1.540562 (A)

46-1212 Quality: *

CAS Number:

Molecular Weight: 101.96
Volume[CD]: 254.81
Dx: 3.987 Dm:

Sys: Hexagonal
Lattice: Rhomb-centered
S.G.: R3c (167)
Cell Parameters:
a 4.758 b c 12.99
α β γ

SS/FOM: F25=358(.0028, 25)
I/Icor:
Rad: CuKa1
Lambda: 1.540562
Filter:
d-sp: diffractometer

Mineral Name:
Corundum, syn

α-Al2 O3
Aluminum Oxide
Ref: Huang, T et al., Adv. X-Ray Anal., 33, 295 (1990)

Fixed Slit Intensity ->
Sqrt Intensity

1.0990

5.9 3.0 2.0 1.5 1.3 d (Å)

d(A)	Int-f	h	k	l	d(A)	Int-f	h	k	l	d(A)	Int-f	h	k	l
3.4797	45	0	1	2	1.5150	2	1	2	2	1.1897	2	2	2	0
2.5508	100	1	0	4	1.5110	14	0	1	8	1.1600	1	3	0	6
2.3794	21	1	1	0	1.4045	23	2	1	4	1.1472	3	2	2	3
2.1654	2	0	0	6	1.3737	27	3	0	0	1.1386	<1	1	3	1
2.0853	66	1	1	3	1.3359	1	1	2	5	1.1256	2	3	1	2
1.9643	1	2	0	2	1.2755	2	2	0	8	1.1241	3	1	2	8
1.7400	34	0	2	4	1.2391	29	1	0	10	1.0990	9	0	2	10
1.6015	89	1	1	6	1.2343	12	1	1	9					
1.5466	1	2	1	1	1.1931	1	2	1	7					

Peak Positions

Bragg's Law: $\lambda = 2d_{hkl} \sin \theta_{hkl}$

The interplanar spacing, d, for a given hkl reflection is given by the unit cell dimensions

Cubic:

$1/d^2 = (h^2 + k^2 + l^2)/a^2$

Tetragonal:

$1/d^2 = \{(h^2 + k^2)/a^2\} + (l^2/c^2)$

Orthorhombic:

$1/d^2 = (h^2/a^2) + (k^2/b^2) + (l^2/c^2)$

Hexagonal:

$1/d^2 = (4/3)\{(h^2 + hk + k^2)/a^2\} + (l^2/c^2)$

Monoclinic:

$1/d^2 = (1/\sin^2 \beta)\{(h^2/a^2) + (k^2 \sin^2 \beta/b^2) + (l^2/c^2) - (2hl\cos \beta/ac)\}$

Example: $SrTiO_3$

The crystal structure of $SrTiO_3$ is cubic, space group *Pm3m* with a unit cell edge a = 3.90 Å. Calculate the expected 2θ positions of the first three peaks in the diffraction pattern, if the radiation is Cu Kα (λ = 1.54 Å).

1. **Recognize the hkl values for the first few peaks:**
   100, 110, 111, 200, 210, 211, 220, etc.
2. **Calculate the interplanar spacing, d, for each peak:**
   $1/d^2 = (h^2 + k^2 + l^2)/a^2$
3. **Use Bragg's Law to determine the 2θ value:**
   $\lambda = 2d_{hkl} \sin \theta_{hkl}$

Example: $SrTiO_3$

***hkl* = 100**

$1/d^2 = (1^2 + 0^2 + 0^2)/(3.90\ Å)^2 \rightarrow d = 3.90\ Å$

$\sin \theta_{100} = 1.54\ Å/\{2(3.90\ Å)\} \rightarrow \theta = 11.4°\ (2\theta = 22.8°)$

***hkl* = 110**

$1/d^2 = (1^2 + 1^2 + 0^2)/(3.90\ Å)^2 \rightarrow d = 2.76\ Å$

$\sin \theta_{100} = 1.54\ Å/\{2(2.76\ Å)\} \rightarrow \theta = 16.2°\ (2\theta = 32.4°)$

***hkl* = 111**

$1/d^2 = (1^2 + 1^2 + 1^2)/(3.90\ Å)^2 \rightarrow d = 2.25\ Å$

$\sin \theta_{100} = 1.54\ Å/\{2(2.25\ Å)\} \rightarrow \theta = 20.0°\ (2\theta = 40.0°)$

Effect of Sample Height Displacement

The most common type of experimental error is a shift of the peaks due to the sample height being slightly shifted from the center of the diffraction circle. The dependence of this displacement is given by the following equation:

$$\Delta 2\theta \text{ (in radians)} = (2s \cos \theta)/R$$

S = sample height displacement

R = Diffractometer radius

Note that the sample height displacement causes a larger error for low angle peaks. Therefore, for the most accurate unit cell parameters it is generally better to use the high angle peaks for this calculation.

Sample Height Displacement: Example

Imagine that you measured the position of the (100) peak and the (800) peak for a cubic phase with a cell edge of 12.00 A, and used these peak positions to calculate the length of the cell edge. How much error would be introduced if the sample was displaced by 100 microns (0.1 mm)? Assume λ = 1.54 A and R = 225 mm.

In the absence of a displacement the locations of these two peaks would be

$d(100) = 12.00\ A \rightarrow \theta = \sin^{-1}\{\lambda/2d\} = \sin^{-1}\{1.54\ A/(2 \times 12.00\ A)\} = 3.68°$

$d(800) = 1.50\ A \rightarrow \theta = \sin\text{-}1\{\lambda/2d\} = \sin\text{-}1\{1.54\ A/(2 \times 1.50\ A)\} = 30.89°$

The displacement will cause each of these peaks to shift by the following amount (assume the diffractometer radius is R = 225 mm):

$\Delta 2\theta = (180/\pi)(2s \cos\theta)/R = (180/\pi)\{2(0.1) \cos 3.68\}/(225) = 0.051°$

$\Delta 2\theta = (180/\pi)(2s \cos\theta)/R = (180/\pi)\{2(0.1) \cos 30.89\}/(225) = 0.044°$

We can add these values to the θ values calculated in the first part and the new peak positions are 7.41° 2q and 61.82° 2q. These give cell edges of:

$a = d(100) = \lambda/\{2 \sin(2\theta/2)\} = 1.54/\{2 \sin(7.41/2)\} = 11.91\ A$

$a = 8 \times d(800) = 8(1.54/\{2 \sin(61.82/2)\}) = 11.99\ A$

Indexing

- Indexing is the process of determining the unit cell dimensions from the peak positions. To index a powder diffraction pattern it is necessary to **assign Miller indices, *hkl*, to each peak**.
- A diffraction pattern cannot be analyzed until it has been indexed. **It is always the first step in analysis**.
- Unfortunately it is not just the simple reverse of calculating peak positions from the unit cell dimensions and wavelength.
- We will show how one can manually index diffraction patterns of high symmetry structures. For lower symmetry structures (orthorhombic, monoclinic, triclinic) it is usually necessary to use a computer algorithm. **This is called Autoindexing**.

Indexing a Cubic Pattern

Bragg's Law tells us the location of a peak with indices hkl, θ_{hkl}, is related to the interplanar spacing, d, as follows:

$$\lambda = 2d_{hkl} \sin \theta_{hkl}$$

$$1/d = 2 \sin \theta / \lambda$$

$$1/d^2 = 4 \sin^2 \theta / \lambda^2$$

Earlier we saw that for a cubic phase the d-values can be calculated from the Miller indices (hkl)

$$1/d^2 = (h^2 + k^2 + l^2)/a^2$$

Combining these two equations we get the following relationship

$$\sin^2 \theta / (h^2 + k^2 + l^2) = \lambda^2 / 4a^2$$

Need to find values of h,k,l for that give a constant when divided by each $\sin^2 \theta$.

Cubic Example

2-Theta	1000 $\sin^2\theta$	1000 $\sin^2\theta$ /CF	hkl
22.21	37.1		
31.61	74.2		
38.97	111		
45.31	148		
51.01	185		
56.29	222		
66.00	297		
70.58	334		
75.03	371		
79.39	408		

We need to find a common factor, CF, that can be divided into each of the values in the second column to give an integer = $h^2+k^2+l^2$

CF = 37.1

2-Theta	1000 $\sin^2\theta$	1000 $\sin^2\theta$ /CF	hkl
22.21	37.1	1.00	
31.61	74.2	2.00	
38.97	111	2.99	
45.31	148	3.99	
51.01	185	4.99	
56.29	222	5.98	
66.00	297	8.01	
70.58	334	9.00	
75.03	371	10.00	
79.39	408	11.00	

CF = (37.1/1000) = $\lambda^2/4a^2 \rightarrow$ (a = 4.00 A)

2-Theta	1000 $\sin^2\theta$	1000 $\sin^2\theta$ /CF	hkl
22.21	37.1	1.00	1 0 0
31.61	74.2	2.00	1 1 0
38.97	111	2.99	1 1 1
45.31	148	3.99	2 0 0
51.01	185	4.99	2 1 0
56.29	222	5.98	2 1 1
66.00	297	8.01	2 2 0
70.58	334	9.00	300/221
75.03	371	10.00	3 1 0
79.39	408	11.00	3 1 1

Systematic Absences - Centering

If the lattice is not primitive certain classes of hkl peaks will be missing. These are called systematic absences and we can use them to determine the space group (or at least narrow down the possibilities). We will derive this relationship next lecture, but consider that if you have a centered cell it is always possible to draw a smaller primitive cell, and a smaller cell should have fewer peaks.

Centering	**Allowed peaks**
I-centered	→ Peaks where h+k+l is an even number
F-centered	→ Peaks where hkl are either all even #s or all odd #s
C-centered	→ Peaks where h+k is an even number
B-centered	→ Peaks where h+l is an even number
A-centered	→ Peaks where k+l is an even number
R-centered	→ Peaks where -h+k+l is a multiple of 3

Systematic Absences – Screws & Glides

Screw axes and glide planes also have elements of translation and they will give systematic absences as well. Some examples are given below, others can be deduced from the pattern.

Centering	**Allowed peaks**
2_1 screw axis || to a	→ h00 peaks are only allowed when h is an even #
2_1 screw axis || to b	→ 0k0 peaks are only allowed when k is an even #
2_1 screw axis || to c	→ 00l peaks are only allowed when l is an even #
3_1 screw axis || to c	→ 00l peaks are only allowed when l = 3n (n = integer)
4_1 screw axis || to c	→ 00l peaks are only allowed when l = 4n (n = integer)

a glide plane ⊥ c → hk0 peaks are only allowed when h is an even #
b glide plane ⊥ c → hk0 peaks are only allowed when k is an even #
n glide plane ⊥ c → hk0 peaks are only allowed when h+k is an even #
a glide plane ⊥ b → h0l peaks are only allowed when h is an even #
c glide plane ⊥ b → h0l peaks are only allowed when l is an even #
n glide plane ⊥ b → h0l peaks are only allowed when h+l is an even #

Indexing & Systematic Absences

Consider the following example of indexing a cubic pattern where there are systematic absences.

Assume Cu radiation, λ = 1.5406 A.

2-theta	d	$1000/d^2$
28.077	3.175	99.2
32.533	2.750	132.2
46.672	1.945	264.3
55.355	1.658	363.8
58.045	1.588	396.6
68.140	1.375	528.9
75.247	1.262	627.9
77.559	1.230	661.0

It's immediately clear that 99.2 is not a common factor here. Though we can see that 132.2-99.2=33 might be a common factor. So we'll give it a try.

2-theta	d	$1000/d^2$		hkl
28.077	3.175	99.2	99.2/33=3	111
32.533	2.750	132.2	132.2/33=4	200
46.672	1.945	264.3	264.3/33=8	220
55.355	1.658	363.8	363.8/33=11	311
58.045	1.588	396.6	396.6/33=12	222
68.140	1.375	528.9	528.9/33=16	400
75.247	1.262	627.9	627.9/33=19	331
77.559	1.230	661.0	661.0/33=20	420

From the absences we see that the compound is F-centered, a = $[1000/33]^{1/2}$ = 5.50 Angstroms

Autoindexing

- Manual indexing of cubic unit cells is a reasonably straightforward process.
- Tetragonal, trigonal and hexagonal cells can also be indexed manually with some experience, but it is not a trivial exercise.
- Generally indexing is done using a computer program. This process is called autoindexing.
- The input for an autoindexing program is typically:
  - The peak positions (ideally 20-30 lines)
  - The wavelength
  - The uncertainty in the peak positions
  - The maximum allowable unit cell volume

Autoindexing Software

A number of the most useful autoindexing programs have been gathered together by Robin Shirley into a single package called **Crysfire**. You can download Crysfire from the web and find tutorials on its use at

http://www.ccp14.ac.uk/tutorial/crys/index.html

To go index a powder diffraction pattern try the following steps:

- Fit the peaks using a program such as X-Fit (http://www.ccp14.ac.uk/tutorial/xfit-95/xfit.htm)
- Take the X-fit output file and convert to a Crysfire input file, as described on the web.
- Run Crysfire to look for the best solutions.
- Evaluate the systematic absences and refine the cell parameters. This can be done using the material in the front of the international tables for crystallography or using a program like Chekcell (http://www.ccp14.ac.uk/tutorial/lmgp/index.html).

Autoindexing - Pitfalls

- **Inaccurate data**
  - Analytically fit the peaks
  - Either correct for or avoid sample displacement error (internal standard if necessary)
- **Impurities**
  - Try different programs
  - Drop out various weak peaks
  - Try different sample preps
  - Complimentary analysis
- **Psuedosymmetry**
  - Unit cell dimensions are close to a more symmetric crystal system
- **Inadequate number of peaks**
  - You really need 15-25 peaks, particularly if the symmetry is low

How do I know when I'm finished?

- Evaluate output based on figure of merit, when the following conditions are met the solution warrants close consideration
  - $M_{20} > 10$
  - All of the peaks are indexed
- Solutions with figures of merit above 20 or so almost always have some degree of the truth in them, but closely related solutions and partially correct solutions are common.
- Favor high symmetry solutions over low symmetry ones.
- Autoindexing is a way to get your foot in the door. Solutions always have to be checked further.

References

- "Structure and Bonding in Crystalline Materials", G.S. Rohrer, Chapter 4, Cambridge University Press, Cambridge (2001).
- "Elements of X-ray Diffraction" B.D. Cullity, Addison-Wesley Pub. Co., (1978).
- "Autoindexing" by P.E. Werner, Chapter 7 in "Structure Determination from Powder Diffraction Data" IUCr Monograph Vol. 13. Edited by W.I.F. David, K. Shankland, L.B. McCusker, Ch. Baerlocher, Oxford University Press (2002).
- A rather complete website with programs, references and discussion of indexing powder patterns can be found at http://www.ccp14.ac.uk/solution/indexing/index.html